

Final Project Report

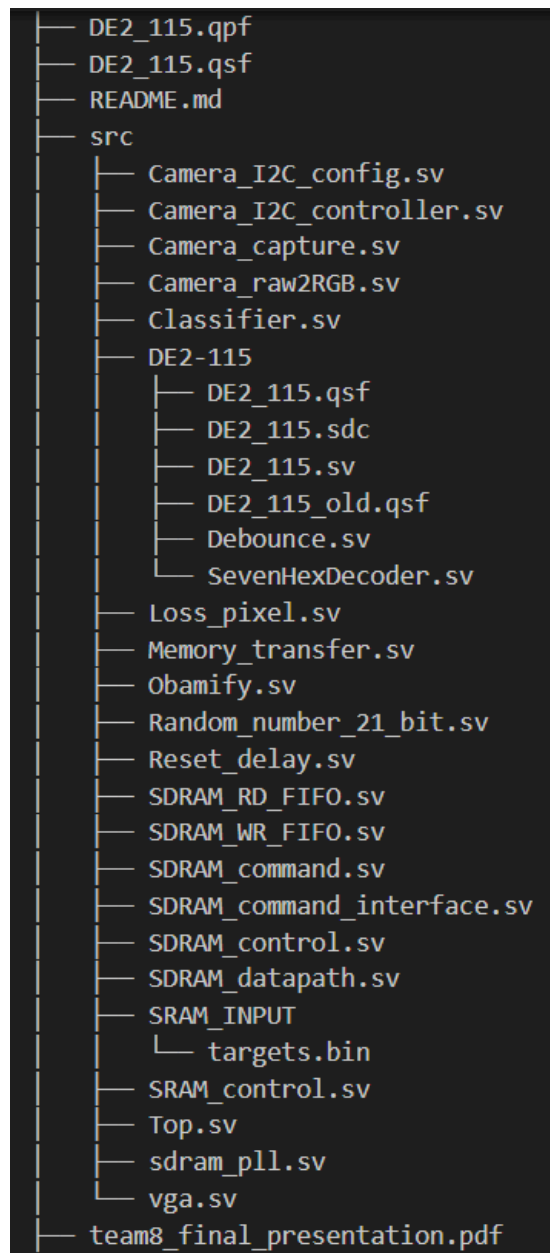
b11901013俞柏安

b11901042連子力

b11901054吳柏均

File Structure

The below image shows the overall file structure.



System Architecture

Off-board peripheral:

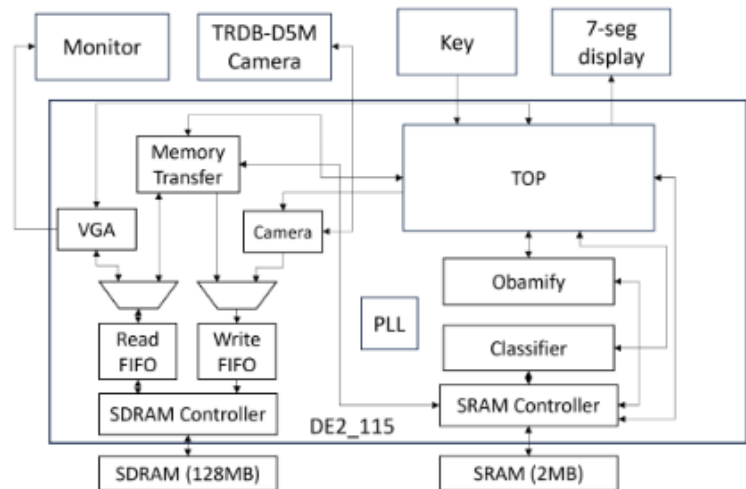
- monitor
- camera

On-board peripheral:

- key
- 7 seg display
- SDRAM
- SRAM

Use PLL to provide clock for vga, SRAM, SDRAM, etc.

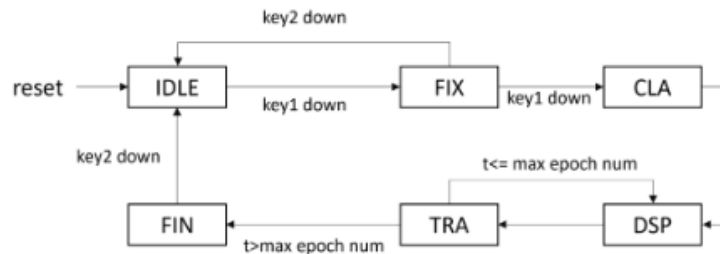
Top: state machine and distribute control signal



Hardware Scheduling (FSM or Algorithm Workflow)

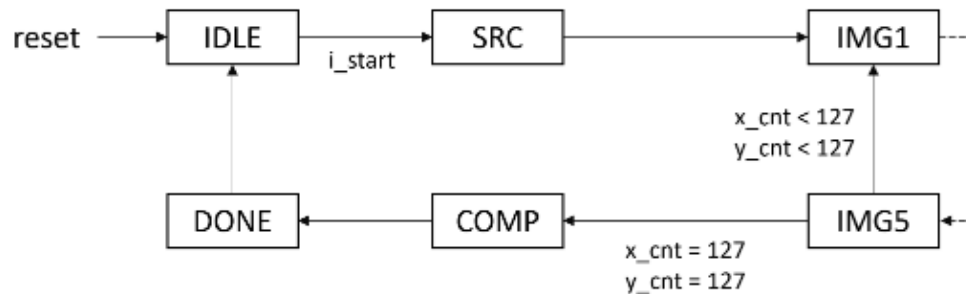
Main States:

- **IDLE**: ready to take photo
- **FIX**: fixed image on screen
- **CLA**: classification
- **DSP**: obamify
- **TRA**: transfer file from SRAM to SDRAM
- **FIN**: fixed post-obamify image



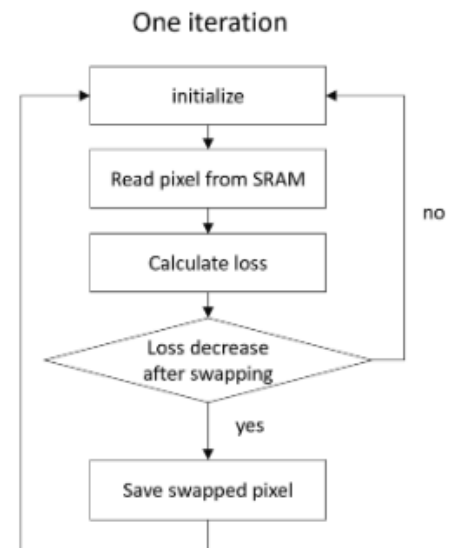
Classification Algorithm:

- **IDLE**: waiting for start
- **SRC**: fetch source image pixel
- **IMG1-5**: fetch source image pixel / calculate SAD (Sum of Absolute Differences)
- **COMP**: compare loss
- **DONE**: output the index of image with minimum index



Obamify (DSP) Algorithm:

- Calculate loss and change the original image if necessary in an iteration
- whole process: M iterations * N epochs
- use a random generator to decide pixel positions
- reading requires 16 cycles, writing requires 4 cycles
- Loss: $(\Delta R^2 + \Delta G^2 + \Delta B^2) / 256$



Fitter Summary

The screenshot shows the Quartus II IDE with the 'Fitter Summary' window open. The 'Table of Contents' pane on the left lists various reports, with 'Fitter Summary' selected. The main pane displays the following information:

Fitter Status
Successful - Fri Dec 26 14:39:55 2025

Quartus II 64-Bit Version
15.0.0 Build 145 04/22/2015 S1 Full Version

Revision Name
DE2_115

Top-level Entity Name
DE2_115

Family
Cyclone IV E

Device
EP4CE115F29C7

Timing Models
Final

Total logic elements
34,415 / 114,480 (30 %)

Total combinational functions
15,327 / 114,480 (13 %)

Dedicated logic registers
29,547 / 114,480 (26 %)

Total registers
29547

Total pins
414 / 529 (78 %)

Total virtual pins
0

Total memory bits
256,128 / 3,981,312 (6 %)

Embedded Multiplier 9-bit elements
12 / 532 (2 %)

Total PLLs
1 / 4 (25 %)

Timing Analyzer

The screenshot shows the Quartus II IDE with the 'Timing Analyzer' window open. The 'Table of Contents' pane on the left lists various reports, with 'Timing Analyzer' selected. The main pane displays the following information:

Unconstrained Paths

	Property	Setup	Hold
1	Illegal Clocks	0	0
2	Unconstrained Clocks	1	1
3	Unconstrained Input Ports	1	1
4	Unconstrained Input Port Paths	4	4
5	Unconstrained Output Ports	60	60
6	Unconstrained Output Port Paths	162	162

Problems Encountered & Solutions

1. Problem: FIFO needs several cycles after reset, causing it to miss data from Camera/VGA module.

Solution: Separate the Reset signal into different timings, reset FIFO first, then reset/start Camera/VGA module.

2. Problem: Camera module returns unexpected results. (e.g. different order)
Solution: Adjust I2C Configurations/receive logic to decode into the correct RGB data.
3. Problem: SDRAM, SDRAM Control, FIFO has unexpected behavior occasionally.
Solution: Added clock source declaration in .sdc file to ensure these modules meet timing requirements during synthesis.
4. Problem: VGA has unexpected behavior, including glitches and glitters
Solution: Output pure color first. Additionally, the signal should be zero in back porch, front porch and sync pulse, or the protocol isn't met and error could occur.
5. Problem: The original 128*128 image is too small, but memory space is limited
Solution: Use zero-order interpolation to render 1 pixel to 4 pixels. While the coordinate is in the specific range, toggle the flag enabling request from SDRAM when x or y increment.
6. Problem: The Obamify algorithm is difficult and hard to debug, so directly using verilog code and FPGA is inefficient.
Solution: We used a Python simulation program to simulate the algorithm, constrained by bit width, cycles, memory usage, etc. This approach allowed us to merge the Obamify algorithm within a short period of time.